Equivalence partitioning:

Equivalance partitioning is the test design technique.

which Input data is devided into groups called partitions.

One representative value from each partition is selected to the test system.

Example;

Age validation (20-60)

valid partition : 20-60.

Invalid partion : 0-19.

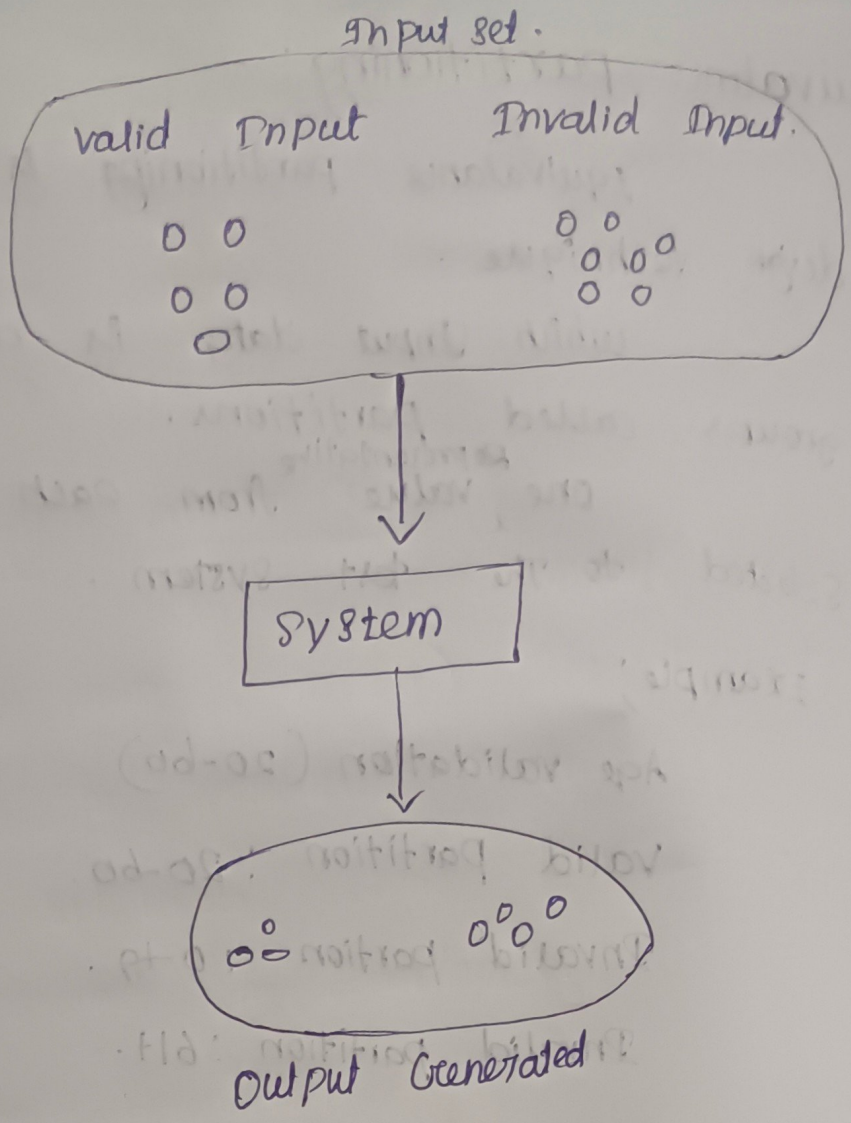
Invalid partition : 61+.

A common Example is testing a form that requires a user's age to be between 20 and 60.

valid partition : Ages. 20-60, A test case could be a User Aged 45.

Invalid partition : Ages 0-19. A test case could be a user aged 15.

Invalid partition : Ages 61 and above. A test case could be a User aged 70.



Guidelines for Equivalence partitioning :

1.) If Input conditions is a range :

When system accepts a range of values,

we create :

1. valid partition.
2. invalid partition.

Example ; Input range 1 to 100 100.

valid partion → 1-100.
Invalid partion → less than 1.
Invalid partion → Greater than 100.

2. If the input condition is a specific value:

If the system accepts only one specific value, than we create;

1. valid partition.
2. Invalid partition.

Example: System accepts only 10.

valid → 10.
Invalid → Less than 10.
Invalid → Greater than 10.

3. If Input condition is a set of values:

If the system accepts a set of values, than we create; 1 valid partion, 1 Invalid partion.

Example; Accepted values → {Red, Blue, Green}
valid → {Red, Blue, Green}.
Invalid → {yellow}

4. If the Input condition is Boolean:

If the input is Boolean, there are two possible values.

valid → True Invalid → false.

2. Boundary Value Analysis :

BVA is a software testing technique used to test the minimum and maximum boundary values of input ranges to detect errors.

Identify bundaries :

for a given input range (eg. age from 18 to 60), The minimum is 18 and the maximum is 60.

Test value : Create test cases using values around the boundaries.

Minimum boundary :

Test The minimum value itself, one value below it, and one value above it (eg. 17, 18, 19)

Maximum Boundary :

Test the maximum value itself, one value below it, and one value about it. (eg. 59, 60, 61).

Test valid and valid inputs :

Ensure the software handles Both valid boundary conditions (like 18

and 60) and invalid boundary conditions (like 17 and 61) correctly.

Focus on edge cases:

~~By testing these Extreme values~~

Edge cases refer to testing the extreme bundary values of input ranges. By testing minimum, maximum, and values just above and below them, errors at the ~~bou~~ boundries can be detected.

~~Points to consider in BVA~~:

valid values : Test with 7, 8, 14 and 15 characters.

Invalid values : Test with 6 characters and 16 characters to ensure the system reject them.